

MZUZU DIOCESE

2021 MALAWI SCHOOL CERTIFICATE OF EDUCATION MOCK EXAMINATION

CHEMISTRY

PAPER I (100 marks)

Theory

Subject number: M162

Wednesday, 4 August

Time Allowed: 2 hours
1:30 – 3:30

Instructions

1. This paper contains 14 printed pages. Please check.
2. Write your name clearly in the space provided on top of this page and all other pages.
3. In Section A short answer questions while in Section B three restricted essay questions.
4. thirteen
5. Use of electronic calculators is allowed.
6. The maximum number of marks for each answer is indicated against each question. This paper has a total of 100 marks .
7. In the table provided on this page, tick against the
8. Hand in your paper to the invigilator when time is called to stop writing.

Question Number	Tick if answered	Do not write these columns
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		

Total

1. Describe any one component of a scientific investigation.

(1 mark)

State any two importance of drawing line graph to interpret data in a table during scientific investigation.

i. _____ (1 mark)

ii. _____ (1 mark)

and T form ions with the electron configuration as shown below.



i. _____ and T. _____ (1 mark)

_____ (1 mark)

T.

i. _____ (1 mark)

ii. _____ (1 mark)

2. a. Explain the effect on atomic radius when valence electrons are lost.

(2 marks)

b.

C₂H₆).



(3 marks)

exothermic reaction '.

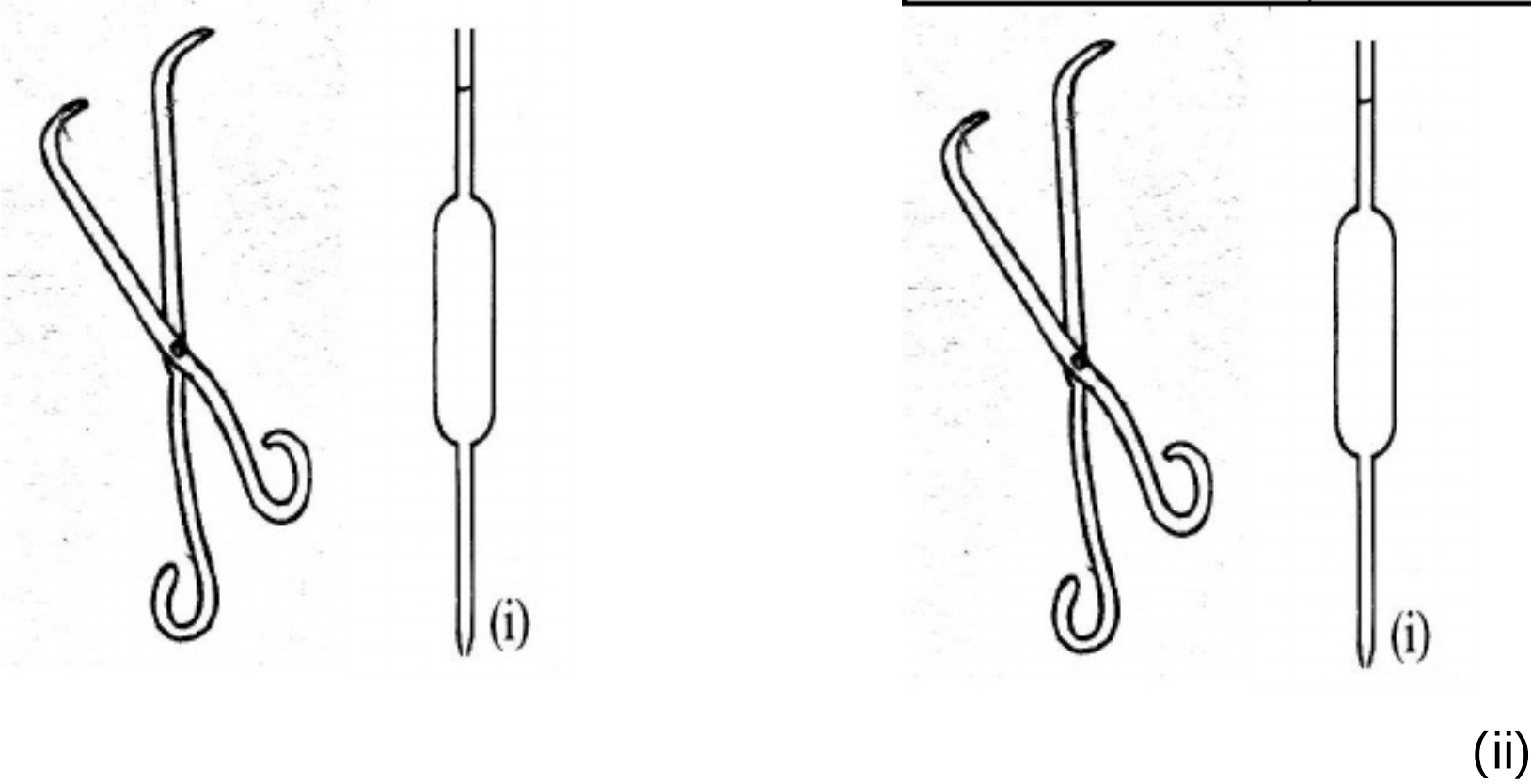
(1 mark)

ii. Construct a well labelled energy level diagram for the process of photosynthesis represented by the following thermo-chemical equation:



(3 marks)

Figure 1 below shows laboratory apparatus (i) and (ii)					



Name the apparatus shown in the Figure above.

(i) _____ (1 mark)

(ii) _____ (1 mark)

c.State the use of apparatus (i)

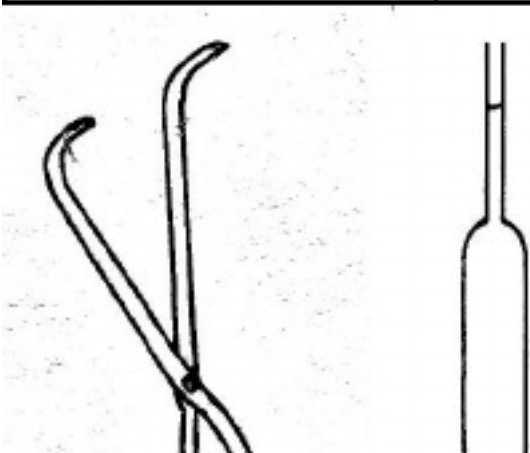
 _____ (1 mark)

4. a. **Table 1** shows results of an experiment on displacement reactions involving zinc (Zn), aluminium (Al), tin (Sn) and iron (Fe) metals and their aqueous solutions.

Table 1				
	Zn $^{2+}$ (aq)	Al $^{3+}$ (aq)	Sn $^{2+}$ (aq)	Fe $^{2+}$ (aq)
				Reaction
				Reaction

No reaction

No reaction



i. Write a balanced chemical equation for the reaction between zinc (Zn) and iron sulphate (FeSO₄) solution.

_____ (2 marks)

ii. Use the results to arrange the metals in order of increasing reactivity.

_____ (1 mark)

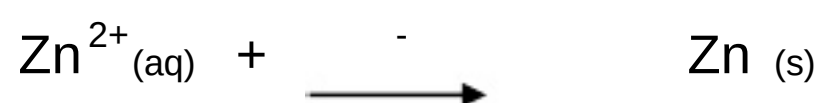
iii. Which of the metals is the strongest reducing agent?

_____ (1 mark)

b. Below are electrode potentials of some half-reaction reactions.

Half -

? volts)



? **0.76**



0.77

Calculate the e.m.f. of the cell.

→

→

(3 marks)

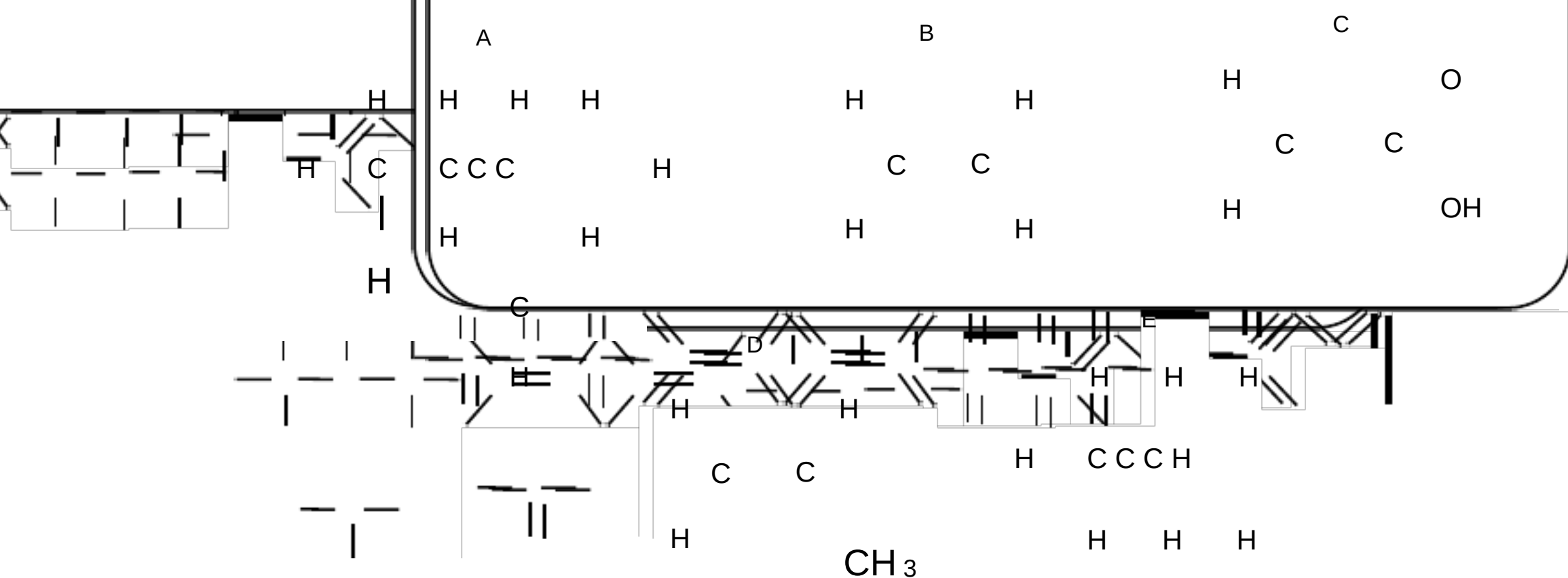
oxidation number ' _____

_____ (1 mark)

- ii. Work out the oxidation number of sulphur (S) in a sulphite ion (SO_3^{2-}) given that oxidation number of oxygen (O) is -2 .

(2 marks)

and E.



Name compound A.

_____ (1 mark)

- ii. Which compound is soluble in water?

_____ (1 mark)

- iii. Give a reason for the answer in 5. a. ii.

 _____ (1 mark)

Compound B is a monomer. Write an equation to show its polymerization.

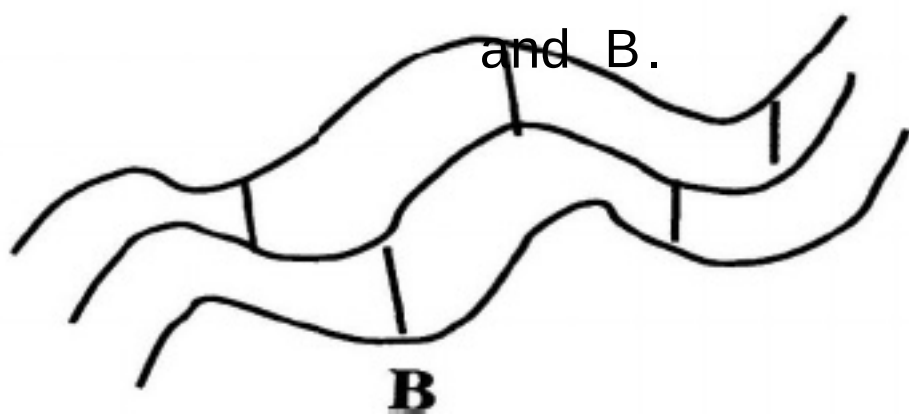
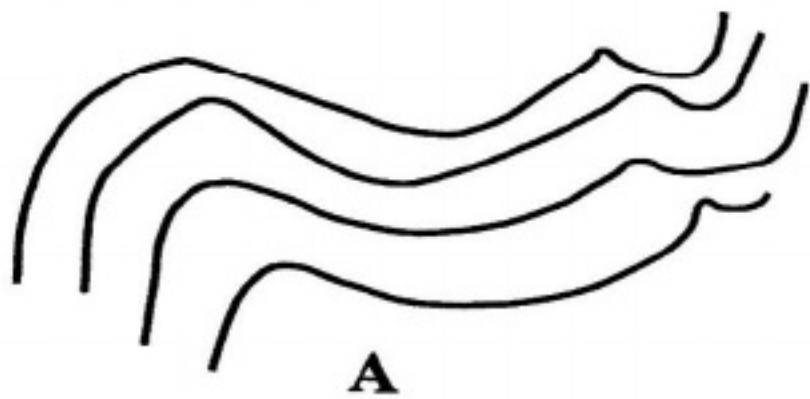
(1 mark)

Mention the kind of polymerization in 5. a. iv.

(1 mark)

vi. Write the other isomers of substance A.

(2 marks)



i. Which structure shows plastic that can be recycled?

(1 mark)

ii. Give a reason for the answer in 5.b. i.

(1 mark)

iii.

that is used to make clothing and ropes .

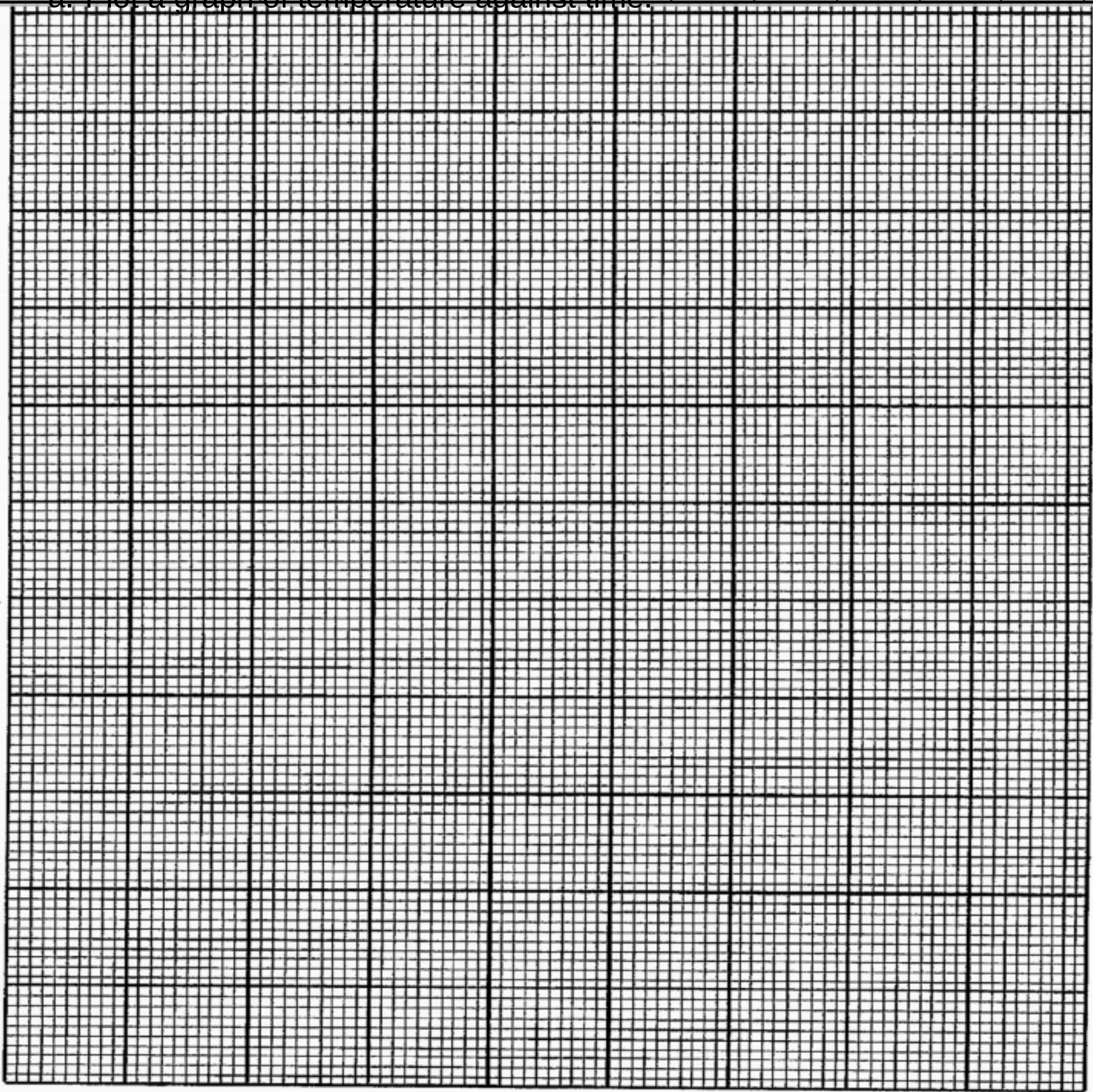
(1 marks)

6. **Table 2** gives readings of a thermometer collected at regular time intervals by a student as an unknown substance was heated from solid state to gaseous state.

Temperature (⁰											

a. Plot a graph of temperature against time.

(4 marks)



b. Use the graph to predict if the unknown substance was pure or impure.

(1 mark)

c. What was the temperature of the substance after 28 seconds?

(1 mark)

d. Why is it necessary to specify the temperature of a substance when investigating its purity using density?

(1 mark)

What is a 'limiting reagent' in a chemical reaction?

(1 mark)

b. Calcium carbonate can be decomposed by heating according to the following equation



on heating 25.9g of Calcium carbonate (CaCO_3)?

(4 marks)

two allotropes of phosphorus.

_____ (2 marks)

Give any two differences between diamond and graphite.

i. _____
_____ (1 mark)

ii. _____
_____ (1 mark)

c. What name is given to the process of manufacturing ammonia gas from nitrogen?
_____ (1 mark)

e. How is nitrogen useful in transporting petroleum products?

_____ (1 mark)

f. State any two uses of sulphur.

i. _____
_____ (1 mark)

ii. _____
_____ (1 mark)

9. a. What is a “weak acid?”

_____ (1 mark)

b. Mention the acid which could be used to prepare ammonium chloride (NH_4Cl).
_____ (1 mark)

c.

i. Reacting an acid with a metal

(1 mark)

ii. Reacting an acid with a carbonate

(1 mark)

d. The conjugate acid-base pairs for the reaction between water molecules are

$\text{H}_2\text{O} / \text{OH}^-$ and $\text{H}_3\text{O}^+ / \text{H}_2\text{O}$. **Write an equation for the reaction.**

_____ (2 marks)

10. State any two ways of expressing concentration of a solution.

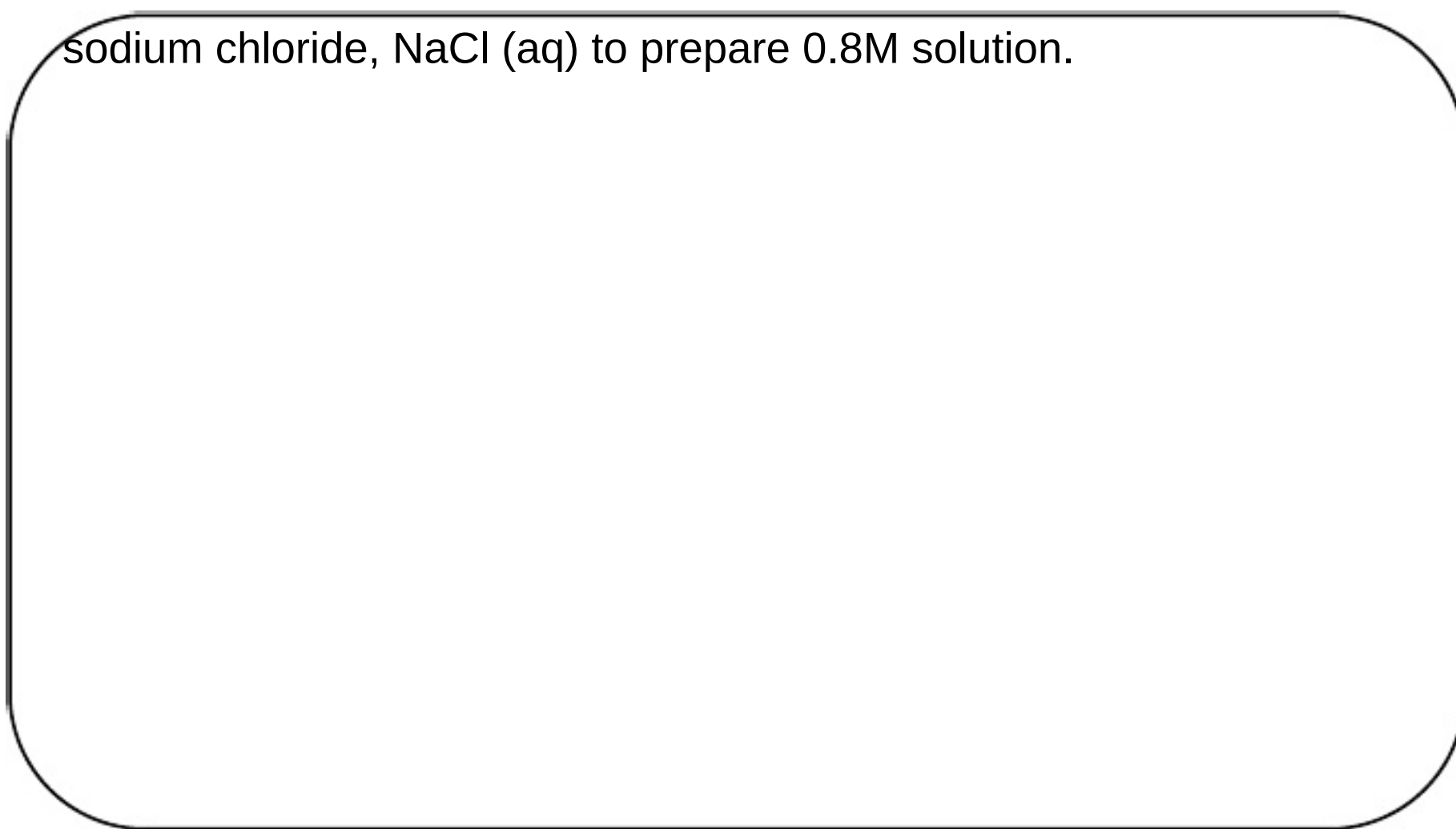
i. _____

(1 mark)

ii. _____

(1 mark)

b. Calculate the volume of distilled water that must be added to 128 ml of 1.25M sodium chloride, NaCl (aq) to prepare 0.8M solution.



(3 marks)

11. Explain any one physical method of removing water hardness.

(2 marks)

b. With the aid of well labelled diagram explain how water circulates continuously between the earth's surface and the atmosphere.

(8 marks)

12. With the aid of a well labelled diagram, describe how alcohol can be produced locally from cereals, sugar and water.

[illegible]

(8 mark

13. Describe how 250cm³ of a 0.5M magnesium sulphate solution could be prepared using hydrated magnesium sulphate crystals (MgSO₄ · 7H₂O).

[illegible]

(10 mar

END OF QUESTION PAPER